

Report: Prototyping Methods for Scalable Small Business Network

1. Cisco Packet Tracer (Simulation)

Explanation:

A network simulation tool from Cisco. You drag and drop virtual routers, switches, PCs, and cables onto a canvas. It simulates how data packets move but does not run real operating system code. Best for learning logic and protocols (VLANs, STP, DHCP, static routing).

Advantages:

- **Very easy to use** – GUI is intuitive; no command-line complexity for basic setups.
- **Fast prototyping** – Build a 20-device network in 30 minutes.
- **Low resource usage** – Runs on any laptop (2GB RAM, no virtualization needed).
- **Built for grading** – Easy to show connectivity (ping success/fail, simulation mode).
- **Free & legal** – No licensing costs for students.

Disadvantages:

- **Not real code** – Some advanced features (specific routing protocols, QoS) behave differently than real hardware.
- **No true OS** – Cannot run actual server applications (web, database, email) inside the simulated PCs.
- **Performance unrealistic** – Does not show real throughput, latency, or CPU load under traffic.
- **Vendor lock-in** – Commands are Cisco-specific; not transferable to MikroTik, Ubiquiti, or open-source firewalls.

2. VirtualBox with Virtual Machines (Emulation)

Explanation:

VirtualBox runs real operating systems (Linux, VyOS, pfSense, Windows) as virtual machines

on your laptop. You create virtual network adapters and connect VMs using internal networks or host-only adapters. Each VM runs actual network stack code. This is *emulation*, not simulation.

Advantages:

- **Real behavior** – Commands work exactly as on physical hardware (., iptables, sysctl, ip route).
- **Full server capabilities** – Run real web servers, DHCP services, or file shares inside VMs.
- **Vendor agnostic** – Use VyOS (Cisco-like), OpenWRT, or pfSense to prove cross-platform knowledge.
- **Scriptable scalability** – Write scripts to clone a new VM, proving "automated growth."
- **Realistic traffic** – Generate actual throughput (limited by laptop CPU/RAM).

Disadvantages:

- **Steep learning curve** – Requires basic Linux command-line and networking (bridges, routing tables, namespaces).
- **Resource heavy** – Each VM needs 256–512MB RAM + 1 CPU core. A 5-VM lab needs 4GB+ RAM easily.
- **Manual complexity** – Setting up 3 switches + 5 PCs as VMs is tedious; virtual switches require bridge-utils or Open vSwitch.
- **No pretty diagrams** – You manage via CLI and maybe VirtualBox GUI; no built-in topology visualizer.

Best for: Your defense if you want to show *realistic scalability through scripting* – e.g., a Python script that provisions a new employee VM and assigns VLAN tagging dynamically.

Quick Comparison Table

Feature	Packet Tracer	VirtualBox + VMs
Type	Simulation	Emulation
Real OS code?	No	Yes
Resource use	Very low	High
Setup time	30 mins	2–4 hours
Shows real throughput?	No	Yes (limited by host)

Scalability demo	Drag/drop new switch	Script to clone VM
Best for defense	Logical design proof	Realistic behavior proof